

A Resource Theory of the Non-Hermitian Skin Effect: Generator-Anchored Objects, Intrinsic Free Operations, Proved Monotones, and an Explicit Status Ledger

V29 working manuscript

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Abstract

V28 declared universality but rested on definitions that did not single out the skin effect: any family of response states with boundary weight qualified, Hermitian systems included, and the “equivalence” between the physical predicate and the resource spectrum was a re-writing of one inequality. V29 replaces that construction. Objects are *ensembles of right/left response states with boundary collars*, anchored to non-Hermitian generators. Free operations are defined intrinsically (a common CPTP map on the classical–quantum ensemble register together with a stochastic relabelling, subject to *collar non-expansion*) and form a symmetric monoidal category. We prove: (i) a *reciprocity no-go*: every object with $\rho^R = \rho^L$ and disjoint dual collars has bounded skin score, so Hermitian and any reciprocal system is excluded; (ii) skin accumulation forces R/L non-normality, but not conversely (with an explicit bound and a counterexample); (iii) an infinite family of monotones (all coordinatewise convex non-increasing functionals of the leakage profile) plus an additive fidelity monotone, and exact rate-level monotonicity; (iv) a canonical free object reachable from every object, and a non-trivial conversion (“dilution”) that lowers the skin rate to any prescribed value; (v) a two-sided, mode-uniform Hatano–Nelson (HN) calibration with explicit constants; (vi) exact and robust one-shot conversion as a branch-free compact-convex problem with a complete dual. **What is not established** is stated in the Status ledger at the end: the equivalence of the physical predicate with the standard spectral definition beyond imported Toeplitz results, a complete monotone family for the full conversion preorder, exact conversion between different HN couplings, and any non-trivial vanishing-error asymptotic rate. The manuscript does not claim these.

1 Scope and corrections to V28

The target remains a universal quantum resource theory of the non-Hermitian skin effect (NHSE). V29 makes “universal” precise as follows: universal over the class $\mathfrak{D}$ of *skin data* (Definition 1) that any finite-dimensional generator with a boundary measurement produces (Definition 2); *not* over “all physical experiments”. The following V28 statements are withdrawn.

- The physical predicate with $\tau = R$ alone (right states near one boundary) is satisfied by a Hermitian system; V28’s “no-go” was therefore not a no-go for its own predicate. V29 uses the *paired* score (Definition 3), for which Theorem 2 holds.
- “Physical NHSE $\Leftrightarrow$ non-trivial resource spectrum” was a re-writing of the identity $\frac{1}{2}[-\log Q]_+ = \Delta$. V29 does not count this as a theorem (Remark 2).

- V28’s “exact rigidity” (Proposition 10) came from a control non-degradation constraint that froze the collar survival probability. V29 has no such constraint; free operations may dilute skin.
- The claim that the POVM embedding is a free isomorphism is dropped; it is an isometric data-reduction only (Proposition 1).
- V28’s branch-wise duality (Theorems 14–15) ignored that a union of convex sets is not convex. V29 removes discrete branches by convexifying label bijections into stochastic kernels (Section 5).
- V28’s tensor-product score additivity used a non-collar event. With the correct product collar (Definition 6) the skin score is *not* additive; only the fidelity monotone is.

2 Objects

Definition 1 (Skin datum). A skin datum X consists of a finite-dimensional Hilbert space H_X ; a finite response set $[M_X]$ with probability weights π^X ; two families of density operators ρ_m^R, ρ_m^L on H_X (right/left responses); and a *collar family*: projectors $\Pi_{\sigma,w}$, $\sigma \in \Sigma = \{-, +\}$, $w \in [W]$, nested in w . Write $\iota(\mp) = \pm$ and $\Lambda_{\sigma,w} = \mathbb{I} - \Pi_{\sigma,w}$. X is *boundary-separated* if $\Pi_{\sigma,w}\Pi_{\iota\sigma,w} = 0$ for all σ, w . The classical–quantum (cq) operators are $\omega_X^h = \sum_m \pi_m |m\rangle\langle m| \otimes \rho_m^h$, $h \in \{R, L\}$.

Definition 2 (Generator datum). A generator datum is a matrix G on $\ell^2(\text{cells}) \otimes \mathbb{C}^d$ (open boundary), with simple spectrum, a cell-position measurement (projective, or a POVM reduced by Proposition 1), right and left eigenvectors R_m, L_m of G normalised to unit vectors, and $\rho_m^R = |R_m\rangle\langle R_m|$, $\rho_m^L = |L_m\rangle\langle L_m|$, uniform weights, and $\Pi_{\pm,w}$ the projectors onto the w outermost cells at the right/left end. This is boundary-separated when $2w \leq \text{number of cells}$. Liouvillian, Floquet and scattering generators enter the same way once a finite matrix with left/right eigen-responses is specified.

Proposition 1 (POVM reduction). *For a finite POVM $\{E_x\}$ on H , $V\psi = \sum_x \sqrt{E_x}\psi \otimes |x\rangle$ is an isometry, $\text{Tr}[(\mathbb{I} \otimes |x\rangle\langle x|)V\rho V^\dagger] = \text{Tr}[E_x\rho]$, and the root fidelity is preserved: $F(V\rho V^\dagger, V\sigma V^\dagger) = F(\rho, \sigma)$. Hence a POVM-based skin datum is replaced by a projective one with identical boundary statistics and identical Φ_Q .*

Proof. $V^\dagger V = \sum_x E_x = \mathbb{I}$. The probability identity is direct. Fidelity is invariant under a common isometry. $\square$

The map $X \mapsto \tilde{X}$ is a data reduction; no claim is made that it is invertible or a free isomorphism.

3 Leakage, the skin score, and the physical predicate

Definition 3 (Paired leakage and score). For response m , boundary label σ and width w ,

$$\lambda_m^R(\sigma, w) = \text{Tr}[\Lambda_{\sigma,w}\rho_m^R], \quad \lambda_m^L(\sigma, w) = \text{Tr}[\Lambda_{\iota\sigma,w}\rho_m^L], \quad \lambda_m^\Sigma = \frac{1}{2}(\lambda_m^R + \lambda_m^L), \quad g_m(\sigma, w) = -\frac{1}{2} \log \lambda_m^\Sigma \in [0, \infty].$$

The right response is tested at boundary σ and the left response at the *dual* boundary $\iota\sigma$. The resource at scale w is $\mathcal{R}_\sigma(w) = \mathbb{E}_\pi g_m(\sigma, w)$.

Definition 4 (Physical directed skin effect). A sequence X_L of skin data exhibits *directed skin accumulation* if there exist σ , widths $w_L \rightarrow \infty$, a scale $a(w) \rightarrow \infty$ and $\kappa > 0$ with $\mathcal{R}_\sigma(w_L) \geq \kappa a(w_L)$ for all large L . It exhibits *extensive* directed skin accumulation if moreover $\pi\{m : g_m \geq \kappa a(w_L)\} \geq f > 0$. Extensive implies directed (as $g \geq 0$, $\mathcal{R} \geq f\kappa a$); the converse fails when a vanishing fraction of responses carries huge scores, which is why both are kept. Rates: $r_\sigma^\pm[a] = \limsup / \liminf \mathcal{R}_\sigma(w_L)/a(w_L)$.

The predicate contains no control experiment and no resource coordinate; the “control” is the intrinsic bound of Theorem 2.

Theorem 2 (Reciprocity no-go). *If $\rho_m^R = \rho_m^L$ for all m and X is boundary-separated, then $g_m(\sigma, w) \leq \frac{1}{2} \log 2$ for all m, σ, w . Hence every reciprocal object, in particular every Hermitian generator, has all rates $r_\sigma^\pm[a] = 0$.*

Proof. $\lambda_m^\Sigma = 1 - \frac{1}{2} \text{Tr}[(\Pi_{\sigma,w} + \Pi_{\iota\sigma,w})\rho_m]$, and $\Pi_{\sigma,w} + \Pi_{\iota\sigma,w}$ is a projector, so $\lambda_m^\Sigma \geq \frac{1}{2}$. $\square$

Theorem 3 (Skin implies non-normality). *Let X be boundary-separated. Then for every m, σ, w*

$$-\log F(\rho_m^R, \rho_m^L) \geq g_m(\sigma, w) - \log 3,$$

and consequently $\Phi_Q := -\log \sum_m \pi_m F(\rho_m^R, \rho_m^L) \geq G_\sigma(w) - \log 3$ with $G_\sigma(w) := -\log \mathbb{E}_\pi e^{-g_m(\sigma, w)}$.

Proof. Measure the three-outcome projective measurement $\{\Pi_{\sigma,w}, \Pi_{\iota\sigma,w}, \mathbb{I} - \Pi_{\sigma,w} - \Pi_{\iota\sigma,w}\}$; root fidelity is non-increasing under measurement, so $F \leq \sum_i \sqrt{p_i q_i}$ with p, q the outcome laws of ρ^R, ρ^L . Write λ^R, λ^L as in Definition 3. Outcome Π_σ : $\sqrt{p_1 q_1} \leq \sqrt{q_1} \leq \sqrt{1 - q_2} = \sqrt{\lambda^L}$. Outcome $\Pi_{\iota\sigma}$: $\sqrt{p_2 q_2} \leq \sqrt{p_2} \leq \sqrt{\lambda^R}$. Third outcome: $p_3 \leq \lambda^R, q_3 \leq \lambda^L$, so $\sqrt{p_3 q_3} \leq \lambda^\Sigma \leq \sqrt{\lambda^\Sigma}$. Since $\sqrt{\lambda^R} + \sqrt{\lambda^L} \leq 2\sqrt{\lambda^\Sigma}$, $F \leq 3\sqrt{\lambda^\Sigma} = 3e^{-g}$. The aggregate bound follows from $\sum_m \pi_m F_m \leq 3\mathbb{E}e^{-g_m}$. $\square$

Remark 1 (The converse is false). Non-normality without skin exists: a Hermitian chain with a strong non-reciprocal bulk defect has $\rho^R \neq \rho^L$ and $\Phi_Q > 0$ but g_m bounded (check V29-06). Thus the theory separates a *skin sector* from a broader *non-normality sector*; Theorem 3 is the one-way bridge.

Proposition 4 (Hatano–Nelson calibration, explicit). *Let G be the open HN chain on L sites with $G_{n+1,n} = e^g, G_{n,n+1} = e^{-g}, g > 0$, boundary label $\sigma = +$ (right end), width $1 \leq w \leq L - 2$. Then $G = SH_0 S^{-1}$ with $S = \text{diag}(e^{gn})$ and H_0 the symmetric hopping matrix, so for mode j , with $k = \pi j/(L + 1)$,*

$$|R_j(n)|^2 \propto e^{2g(n-1)} \sin^2(kn), \quad |L_j(n)|^2 \propto e^{-2g(n-1)} \sin^2(kn),$$

and uniformly in j ,

$$|g_j(+, w) - gw| \leq \log(L + 1) + g + \frac{1}{2} \log \frac{1}{1 - e^{-2g}}.$$

Hence for widths with $w_L/\log L \rightarrow \infty$ and $a(w) = w$: $r_+^-[w] = r_+^+[w] = g$, and the sequence has extensive directed skin accumulation with $f = 1$.

Proof. Similarity gives the laws. Let $Z_R = \sum_n e^{2g(n-1)} \sin^2(kn)$. Since $\sin(kL)^2 = \sin^2 k \geq 4/(L + 1)^2$ (as $k \in [\pi/(L + 1), \pi L/(L + 1)]$), $Z_R \geq e^{2g(L-1)} \cdot 4/(L + 1)^2$, while $\sum_{n \leq L-w} e^{2g(n-1)} \leq e^{2g(L-w-1)}/(1 - e^{-2g})$. So $\lambda^R \leq e^{-2gw}(L + 1)^2/[4(1 - e^{-2g})]$; the same bound holds for λ^L (mirror),

whence $\lambda^\Sigma \leq \max$ and the lower bound on g . For the upper bound, $Z_R \leq e^{2g(L-1)}/(1 - e^{-2g})$ and $p(L - w) + p(L - w - 1) \geq e^{-2g(w+1)}(1 - e^{-2g})[\sin^2(kn) + \sin^2(k(n+1))]$ with $n = L - w - 1$; the bracket equals $1 - \cos k \cos(2kn + k) \geq \sin^2 k/2 \geq 2/(L+1)^2$. Then $\lambda^\Sigma \geq \frac{1}{2}\lambda^R \geq e^{-2g(w+1)}(1 - e^{-2g})/(L+1)^2$. $\square$

The condition $w_L/\log L \rightarrow \infty$ is an honest extra hypothesis: at $w = O(\log L)$ the $O(\log L)$ error is of the size of the signal.

Remark 2 (Spectrum and the tautology). Taking $a = \mathcal{R}_\sigma(w_L)$ gives rate 1 whenever $\mathcal{R} \rightarrow \infty$. So the all-scale spectrum $\{(a, r^-, r^+)\}$ carries exactly the information of the *growth class* of the single function $w \mapsto \mathcal{R}_\sigma(w)$ (exponential, stretched, logarithmic, ...). “Directed skin accumulation $\Leftrightarrow$ non-trivial spectrum” is therefore true by construction (Definition 4), and it is *not* listed as a theorem. The theorem content is: Theorem 2, Theorem 3, Proposition 4, the monotones below, and the imported link to spectral winding (Section 7).

4 Free operations, category, monotones

Definition 5 (Free morphism). A free morphism $\Theta = (\Xi, \kappa) : X \rightarrow Y$ consists of a CPTP map $\Xi : B(\mathbb{C}^{M_X} \otimes H_X) \rightarrow B(\mathbb{C}^{M_Y} \otimes H_Y)$ and a column-stochastic $\kappa \in [0, 1]^{M_Y \times M_X}$ such that

- (M1) $\Xi(\omega_X^h) = \omega_Y^h$ for $h = R, L$;
- (M2) $\Xi^\dagger(|m'\rangle\langle m'| \otimes \mathbb{K}) = \sum_m \kappa_{m'm} |m\rangle\langle m| \otimes \mathbb{K}$ for all m' ;
- (M3) (collar non-expansion) $0 \leq \Xi^\dagger(|m'\rangle\langle m'| \otimes \Pi_{\sigma,w}^Y) \leq \sum_m \kappa_{m'm} |m\rangle\langle m| \otimes \Pi_{\sigma,w}^X$ for all m', σ, w .

(M1) and (M2) give $\pi^Y = \kappa \pi^X$. (M3) says: weight entering a target collar can only come from the same collar of the source; κ is the randomised relabelling of responses. Label bijections are the extreme points of κ .

Lemma 5 (Leakage domination). *For every free morphism, all m' with $\pi'_{m'} > 0$, $h \in \{R, L\}$, σ, w :*
 $\lambda_{m'}^{h,Y}(\sigma, w) \geq \sum_m \beta_{mm'} \lambda_m^{h,X}(\sigma, w)$ *with $\beta_{mm'} = \kappa_{m'm} \pi_m / \pi'_{m'}$ a probability vector in m .*

Proof. By (M2),(M3), $\Xi^\dagger(|m'\rangle\langle m'| \otimes \Lambda^Y) = \Xi^\dagger(|m'\rangle\langle m'| \otimes \mathbb{K}) - \Xi^\dagger(|m'\rangle\langle m'| \otimes \Pi^Y) \geq \sum_m \kappa_{m'm} |m\rangle\langle m| \otimes \Lambda^X$. Take the trace against ω_X^h and use (M1); apply with $\iota\sigma$ for $h = L$. $\square$

Proposition 6 (Symmetric monoidal category). *Skin data and free morphisms form a category (identity = $(\text{id}, \mathbb{K})$). With the product below it is symmetric monoidal.*

Definition 6 (Tensor product). $X \otimes Y$ has $H_X \otimes H_Y$, responses (m, n) with weights $\pi_m \pi_n$, states $\rho_m^h \otimes \rho_n^h$, and *intersection collars* $\Pi_{\sigma,w}^{X \otimes Y} = \Pi_{\sigma,w}^X \otimes \Pi_{\sigma,w}^Y$ (both subsystems in the collar). Unit: $\mathbb{C}$, one response, $\Pi_{\sigma,w} = 1$.

Proof of Proposition 6. Composition: for $\Theta_1 : X \rightarrow Y$, $\Theta_2 : Y \rightarrow Z$, (M1),(M2) compose with $\kappa = \kappa_2 \kappa_1$; for (M3), positivity of $\Xi_1^\dagger$ gives $\Xi_1^\dagger \Xi_2^\dagger(|m''\rangle\langle m''| \otimes \Pi^Z) \leq \sum_{m'} \kappa_{m''m'}^{(2)} \Xi_1^\dagger(|m'\rangle\langle m'| \otimes \Pi^Y) \leq \sum_m (\kappa_2 \kappa_1)_{m''m} |m\rangle\langle m| \otimes \Pi^X$. Tensor: $\Xi_1 \otimes \Xi_2$ and $\kappa_1 \otimes \kappa_2$ satisfy (M1),(M2) directly. For (M3), $0 \leq A \leq A'$, $0 \leq B \leq B'$ imply $A \otimes B \leq A' \otimes B'$, applied to $A = \Xi_1^\dagger(\cdot \otimes \Pi^Y)$, etc. Associators and swap are the standard unitaries and commute with (M1)–(M3). The unit acts trivially. Boundary-separated data are closed under $\otimes$ (products of orthogonal projectors are orthogonal) but do not contain the unit. $\square$

Remark 3. The unit has $\lambda = 0$, i.e. infinite score: it is the *top* element (there is a morphism $\mathbb{K} \rightarrow X$ for every X and none back). It is not a free object. This is a feature of intersection collars and is stated rather than hidden.

Definition 7 (Free objects). $\mathcal{F}$ is the class of boundary-separated skin data with $\rho_m^R = \rho_m^L$ for all m (reciprocal).

Theorem 7 (Monotones). *Let $\Theta : X \rightarrow Y$ be free.*

1. *For every coordinatewise non-increasing convex lower semicontinuous $\psi : [0, 1]^d \rightarrow (-\infty, \infty]$ applied to the leakage vector $\Lambda_m = (\lambda_m^R(\sigma, w), \lambda_m^L(\sigma, w))_{\sigma, w}$: $\Psi_\psi(Y) := \sum_{m'} \pi'_{m'} \psi(\Lambda'_{m'}) \leq \Psi_\psi(X)$. In particular $\mathcal{R}_\sigma(w)$, $G_\sigma(w)$ and $\sup_{w' \leq w}$, $\sup_\sigma$ of these are non-increasing, and all rates $r_\sigma^\pm[a]$ are non-increasing.*
2. *Φ_Q is non-increasing under every CPTP Ξ satisfying (M1) (no collar condition needed), and $\Phi_Q(X \otimes Y) = \Phi_Q(X) + \Phi_Q(Y)$.*
3. *Free objects have $\Phi_Q = 0$ and all rates 0; hence if $X \in \mathcal{F}$ and $X \rightarrow Y$ then Y has all rates 0.*

Proof. (1) By Lemma 5 and monotonicity then Jensen, $\psi(\Lambda'_{m'}) \leq \psi(\sum_m \beta_{mm'} \Lambda_m) \leq \sum_m \beta_{mm'} \psi(\Lambda_m)$. Multiplying by $\pi'_{m'}$ and summing, $\sum_{m'} \pi'_{m'} \pi_m = \pi_m$ (column-stochasticity) gives $\Psi_\psi(Y) \leq \Psi_\psi(X)$. The score $g = -\frac{1}{2} \log \frac{1}{2}(\lambda^R + \lambda^L)$ is a convex non-increasing function of Λ , and $e^{-g} = \sqrt{\lambda^\Sigma}$ has $-\sqrt{\cdot}$ convex non-increasing, so G_σ is monotone. Rate monotonicity: the inequality holds for each L and w . (2) For cq states with equal classical part, $F(\omega^R, \omega^L) = \sum_m \pi_m F(\rho_m^R, \rho_m^L)$. Uhlmann monotonicity under Ξ gives F non-decreasing, hence Φ_Q non-increasing. Additivity: root fidelity is multiplicative on tensor products of ensembles. (3) Theorem 2 and $F(\rho, \rho) = 1$. $\square$

Proposition 8 (Canonical free object and erasure). *Assume a state ω on H_X with $\text{Tr}[\Pi_{\sigma, w}^X \omega] = 0$ for all σ, w (a bulk state). Let $O_X = (H_X, \{1\}, \rho^R = \rho^L = \omega, \text{collars of } X)$. Then $O_X \in \mathcal{F}$ and $\Xi(A) = \text{Tr}(A) |1\rangle\langle 1| \otimes \omega$ with $\kappa = (1, \dots, 1)$ is a free morphism $X \rightarrow O_X$.*

Proof. (M1), (M2) are direct. (M3): $\Xi^\dagger(|1\rangle\langle 1| \otimes \Pi) = \text{Tr}[\Pi \omega] \mathbb{K} = 0$, and $0 \leq 0 \leq \sum_m |m\rangle\langle m| \otimes \Pi^X$. $\square$

Proposition 9 (Dilution: non-trivial conversion lowering the rate). *For $t \in [0, 1]$, let X_t have responses $\rho_m^h \mapsto (1-t)\rho_m^h + t\omega$ (bulk ω as above), same weights and collars. Then $\Xi_t = (1-t)\text{id} + t\Xi_{\text{er}}$, with $\Xi_{\text{er}}(A) = \sum_m |m\rangle\langle m| \otimes \text{Tr}[(\langle m| \otimes \mathbb{K}) A (|m\rangle \otimes \mathbb{K})] \omega$ and $\kappa = \mathbb{K}$, is a free morphism $X \rightarrow X_t$, and $\lambda_m^\Sigma(X_t) = (1-t)\lambda_m^\Sigma(X) + t$. For $t \leq \frac{1}{2}$,*

$$|g_m(X_t) - \min\{g_m(X), \frac{1}{2} \log(1/t)\}| \leq \frac{1}{2} \log 2.$$

Consequently, with $t_L = e^{-2ca(w_L)}$, $r_\sigma^\pm[a](X_t) \leq \min\{r_\sigma^\pm[a](X), c\}$, with equality when $g_m/a \rightarrow r$ uniformly in m (as for HN).

Proof. (M1), (M2) are direct. (M3): $\Xi_t^\dagger(|m'\rangle\langle m'| \otimes \Pi) = (1-t)|m'\rangle\langle m'| \otimes \Pi \leq |m'\rangle\langle m'| \otimes \Pi$. The band: $(1-t)\lambda + t \leq 2 \max(\lambda, t)$ and $\geq \frac{1}{2} \max(\lambda, t)$ for $t \leq \frac{1}{2}$. $\square$

So every rate below that of X is reachable from X by a free operation, and none above by Theorem 7.

Theorem 10 (Complete monotone family for the leakage preorder). *For finite skin data X, Y the following are equivalent: (i) there is a column-stochastic κ with $\pi^Y = \kappa \pi^X$ and $\Lambda_{m'}^Y \geq \sum_m \beta_{mm'} \Lambda_m^X$ coordinatewise (the conclusion of Lemma 5); (ii) $\Psi_\psi(Y) \leq \Psi_\psi(X)$ for all coordinatewise non-increasing convex ψ .*

Proof. (i) $\Rightarrow$ (ii) is the proof of Theorem 7(1). (ii) $\Rightarrow$ (i) is Strassen’s theorem for the decreasing convex order (Strassen 1965; Müller–Stoyan 2002): (ii) yields a coupling of the two leakage measures with $\mathbb{E}[\Lambda \mid \Lambda'] \leq \Lambda'$, which lifts to a coupling of responses and defines κ . (Imported result; checked numerically in one dimension, V29-07.) $\square$

Remark 4. This is complete for the *leakage preorder*, which is implied by conversion but is coarser than it. A complete single-object monotone family for the full conversion preorder is not established.

5 Exact and robust one-shot conversion

Fix X, Y . Let $\mathcal{K}_0$ be the set of triples $(J, \kappa, \{S_{m'\sigma w}\})$ with J the Choi operator of a CPTP Ξ , κ column-stochastic, and $0 \leq S_{m'\sigma w} \leq \mathbb{K}$. Define the affine residual map

$$\begin{aligned} \mathcal{F}(J, \kappa, S) = & (\Xi(\omega_X^h) - \omega_Y^h \ (h = R, L), \ \Xi^\dagger(|m'\rangle\langle m'| \otimes \mathbb{K}) - \sum_m \kappa_{m'm} |m\rangle\langle m| \otimes \mathbb{K}, \\ & \sum_m \kappa_{m'm} |m\rangle\langle m| \otimes \Pi_{\sigma w}^X - \Xi^\dagger(|m'\rangle\langle m'| \otimes \Pi_{\sigma w}^Y) - S_{m'\sigma w}). \end{aligned}$$

Let $\mathcal{S}_{X \rightarrow Y} = \mathcal{F}(\mathcal{K}_0)$. Slack variables turn the operator inequalities (M3) into equalities; boundedness of S is automatic since $\Xi^\dagger(|m'\rangle\langle m'| \otimes \Pi^Y) \leq \Xi^\dagger(|m'\rangle\langle m'| \otimes \mathbb{K}) \leq \mathbb{K}$.

Lemma 11. $\mathcal{S}_{X \rightarrow Y}$ is compact and convex, and there is no discrete branch: relabellings are absorbed in κ .

Theorem 12 (Exact and robust conversion). (a) $X \rightarrow Y$ iff $0 \in \mathcal{S}_{X \rightarrow Y}$; otherwise a strictly separating affine functional exists. (b) For any norm on the residual space, $\text{dist}(0, \mathcal{S}_{X \rightarrow Y}) = \sup_{\|y\|_* \leq 1} [-\sup_{s \in \mathcal{S}_{X \rightarrow Y}} y \cdot s]_+$.

Proof. (a) is definition-chasing plus finite-dimensional separation of a point from a compact convex set. (b) is the minimax identity $\inf_s \sup_y y \cdot s = \sup_y \inf_s y \cdot s$ on a compact convex set. $\square$

Remark 5. Exact feasibility is an SDP. The theorem is standard convex duality applied to a correctly convexified problem; it is stated for completeness and is not presented as a physical result. The “ ε ” is residual norm: the first block bounds the state error in the chosen norm, the other blocks bound the violation of the free-operation constraints. The witnesses y depend on the pair (X, Y) and are not single-object monotones.

Proposition 13 (What exact conversion between HN objects looks like). *By Theorem 7, $\text{HN}(g) \rightarrow \text{HN}(g')$ requires $g' \leq g$ at the level of rates. In the active orientation, leakage of $\text{HN}(g)$ is pointwise smaller than that of $\text{HN}(g')$ for $g > g' \geq 0$ (monotone likelihood ratio), and in the opposite orientation the order is reversed (V29-08(i)). Since (M3) constrains both orientations, exact conversion $\text{HN}(g) \rightarrow \text{HN}(g')$ is generally obstructed; a numerical feasibility test found it in only 4/24 small cases (V29-08(ii), exploratory). The reachable set from $\text{HN}(g)$ is described by Proposition 9 (dilution), not by a change of coupling. A characterisation of exact HN-to-HN conversion is open.*

6 Tensor powers and asymptotics

Definition 8 (Vanishing-error rate). $R_0(X \rightarrow Y) = \sup\{r : \exists n_k, m_k, \ m_k/n_k \rightarrow r, \ \varepsilon_k \rightarrow 0, \ X^{\otimes n_k} \xrightarrow{\varepsilon_k} Y^{\otimes m_k}\}$, with $\xrightarrow{\varepsilon}$ the residual-norm relation of Theorem 12(b), where the pair (n, m) uses $\mathcal{S}_{X \rightarrow Y}^{(n, m)}$.

By Theorem 12(b), ε -convertibility at (n, m) is $\text{dist}(0, \mathcal{S}_{X \rightarrow Y}^{(n, m)}) \leq \varepsilon$, so “ R_0 is represented by the tensor-power dual witnesses” is a restatement of the definition, and V29 does not present it as a theorem.

Proposition 14 (Zero-error converse). *If $X^{\otimes n} \rightarrow Y^{\otimes m}$ exactly and $\Phi_Q(Y) > 0$, then $m/n \leq \Phi_Q(X)/\Phi_Q(Y)$.*

Proof. Theorem 7(2): $m\Phi_Q(Y) = \Phi_Q(Y^{\otimes m}) \leq \Phi_Q(X^{\otimes n}) = n\Phi_Q(X)$. $\square$

Remark 6 (Why the skin score does not tensorize, and what is open). With intersection collars, $\lambda^R(X^{\otimes n}) = 1 - \mu^n$, so a unit-skin object loses skin only logarithmically in n but does not accumulate it: the skin sector is not additive, whereas Φ_Q is. A vanishing-error converse based on Φ_Q fails as stated: fidelity error ϵ is incompatible with the exponentially small fidelities $e^{-n\Phi_Q}$, so an exponent-level (large-deviation) error criterion is needed. No non-trivial vanishing-error rate is computed here, and no reversibility, equipartition or scalar-charge claim is made.

7 Imported input: spectral winding

The standard definition of NHSE ties boundary accumulation to OBC/PBC spectral sensitivity and point-gap winding. For banded (block-)Toeplitz generators with non-zero winding of the symbol about OBC eigenvalues it is known that a macroscopic fraction of OBC eigenvectors is exponentially localised at a boundary (Okuma–Kawabata–Shiozaki–Sato 2020; Zhang–Yang–Fang 2022; Trefethen–Embree 2005 for the underlying Toeplitz theory). Via Proposition 4-type estimates this places these generators in the directed skin class of Definition 4 with $a = w$. V29 *imports* this and does not re-prove it, and does not claim an equivalence for non-Toeplitz generators (disorder, quasiperiodic, many-body, nonlinear, exceptional-point sectors). For those, only the following holds: if a finite generator datum has right/left eigen-responses, it is a skin datum and Theorems 2–10 apply to it.

8 Numerical support

The companion script `nhse_qrt_v29_checks.py` tests load-bearing statements only; its output is `nhse_qrt_v29_checks_output.txt`. V29-01: reciprocal objects (random mixed states, boundary-concentrated) never exceed $\frac{1}{2} \log 2$ (maximum attained: 0.3466). V29-02: Theorem 3 on random pure and mixed states, minimum slack 0.44. V29-03: 5400 monotone tests over random collar-non-expanding shadow maps with random response mixing, no violation. V29-04: Φ_Q monotonicity under random quantum channels with mixing, and additivity to 10^{-14} . V29-05: analytic HN law agrees with eigen-decomposition (4×10^{-14} at small L); the bound of Proposition 4 holds up to $L = 400$ (numerical eigensolvers are unstable at large L because of pseudospectral sensitivity $\sim e^{gL}$, so the analytic law is used). V29-06: Hermitian and periodic HN controls have bounded score; a bulk non-reciprocal defect has $\Phi_Q > 0$ with bounded score. V29-07: hinge-test equivalence with coupling existence, 400/400. V29-08: leakage order (active/reversed orientation), dilution is a valid free map and satisfies the band, and the exploratory HN→HN feasibility count. **Not tested independently:** Proposition 6 beyond the structure of the proof, Theorem 12 on non-trivial quantum instances, Theorem 10 in dimension $d > 1$, and everything in the Status ledger marked Open or Imported.

9 Conclusion

V29 supplies a resource-theoretic core that is specific to the skin effect: paired boundary leakage of right/left responses, an intrinsic no-go for reciprocal objects, intrinsic collar-non-expanding free operations forming a symmetric monoidal category, a rich proved family of monotones, a canonical free object, a non-trivial conversion, an explicit HN calibration, and a branch-free exact/robust conversion criterion. The universal claim is proved only for the class $\mathfrak{D}$ of skin data and is conditional, through an imported theorem, on the standard spectral definition. The ledger below states the remaining gaps.

References

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Statement	Status	Role
Skin datum and generator datum (Defs. 1,2)	Definition	Fixes the domain $\mathfrak{D}$: generator + boundary measurement + right/left responses. Replaces V28's "all physical experiments".
POVM reduction (Prop. 1)	Proved (elementary)	Removes the projective-cell assumption at the data level; not a free isomorphism.
Reciprocity no-go (Thm. 2)	Proved; V29-01	Intrinsic null sector: Hermitian and all reciprocal objects have zero rate.
Skin $\Rightarrow$ non-normality (Thm. 3)	Proved; V29-02. Converse <i>false</i> (V29-06)	One-way bridge between the skin sector and R/L non-orthogonality.
HN calibration, explicit constants (Prop. 4)	Proved; V29-05	Anchors the exponential sector: $g_j = gw + O(\log L)$ uniformly in j ; needs $w_L / \log L \rightarrow \infty$.
Free morphisms; symmetric monoidal category (Def. 5, Prop. 6)	Proved (closure); structure not independently tested	The arena. Unit is the top element, not a free object.
Monotone family Ψ_ψ, Φ_Q , rate monotonicity (Thm. 7)	Proved; V29-03, V29-04	Resource measures for single objects; Φ_Q is tensor-additive, skin score is not.
Canonical free object and erasure (Prop. 8)	Proved	Free set is reachable from every object.
Dilution lowers any rate to any smaller value (Prop. 9)	Proved; V29-08(iii). Equality of rates only under uniform scaling	First non-trivial conversion; shows the order is not trivial.
Leakage-preorder completeness (Thm. 10)	Imported (Strassen); 1D verified V29-07; $d > 1$ not tested	Complete only for the leakage preorder, which is coarser than conversion.
Exact/robust conversion duality (Thm. 12)	Proved (standard convex duality)	Branch-free, SDP-level criterion. No new physics; witnesses are pair-dependent.
Physical predicate (Def. 4)	Definition; extensive $\Rightarrow$ average proved, converse false	Resource-independent, control-free skin predicate.
"Physical NHSE $\Leftrightarrow$ non-trivial spectrum"	True by construction; <i>not a theorem</i>	Restatement; the spectrum encodes the growth class of $\mathcal{R}_\sigma(w)$.
Link to OBC/PBC sensitivity and point-gap winding (Sec. 7)	Imported for banded Toeplitz; not re-proved. Open beyond Toeplitz	Ties the predicate to the standard NHSE definition. Largest remaining conceptual gap.
Exact conversion $\text{HN}(g) \rightarrow \text{HN}(g')$	Open ; obstructed in tests (4/24 feasible)	Would give an explicit non-trivial order inside the exponential sector.
Complete single-object monotones for the full conversion preorder	Open	Needed to call the ordering fully characterised.
Vanishing-error tensor-power rates	Open ; only zero-error converse (Prop. 14) proved; dual-witness form is definitional	Needed for a genuine asymptotic theory; requires exponent-level error criterion.
Reversible distillation, equipartition, scalar NHSE charge	Not claimed	Outside the present theory.
Universality over all finite boundary experiments	Withdrawn; replaced by universality over $\mathfrak{D}$, conditional on the imported link	States exactly what has been shown.